

DeepHalo Documentation

DeepHalo: A deep learning-integrated workflow for high-throughput discovery of halogenated metabolites from HRMS data.

Version 1.0.0

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Introduction

DeepHalo is a hierarchically optimized workflow designed for the detection and dereplication of halogenated compounds in **high-resolution** mass spectrometry (HRMS) data. It integrates cutting-edge deep learning models, robust isotope validation, comprehensive scoring, and dual dereplication strategies to deliver high accuracy and efficiency. Key applications span multiple fields, ranging from natural product discovery and halogenated metabolite annotation to precise tracking of halogenated drug metabolism in pharmacological studies; from comprehensive monitoring of persistent halogenated pollutants in environmental science to sensitive detection of chlorinated disinfection byproducts in water quality analysis.

Data Support

DeepHalo supports **high-resolution** mass spectrometry (HRMS) data, with or without tandem mass spectrometry (MS/MS) information. Note that the MS² extraction function currently supports only data-dependent acquisition data (DDA). Additionally, DeepHalo

exclusively supports the **.mzML** file format for analysis.

Core Features

1. Halogen Prediction

- **Element Prediction Model (EPM)**
 - Dual-branch Isotope Neural Network (IsoNN) architecture
 - High accuracy Cl/Br detection (>99.9% precision based on benchmark results)
 - Wide mass range coverage (50-2000 Da)
 - Robust interference resistance to B/Se/Fe/dehydro isomers

2. Isotope Pattern Validation

- **Dual Validation System**
 - *Mass Dimension*: Statistical rule-based correction.
 - *Intensity Dimension*: Autoencoder-based Anomaly Detection Model (ADM).

3. Multi-Level Halogen Confidence Scoring (H-score)

- **- Dual levels**
 - Prediction based on centroid-level isotope patterns
 - Prediction based on Scan-level isotope patterns
 - H-score integration for comprehensive assessment on the above both levels

4. Dereplication

- **Dual Strategy Approach**
 - *MS1-based dereplication using Custom Database Matching*: Validates exact mass, halogen patterns, and isotope intensity similarity.
 - *MS2-Based Dereplication by Integrating GNPS*: MS2 molecular networking, halogenated compound annotation, graphML file enhancement.
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Technical Advantages

- **High Throughput**: automatically process unlimited samples in several to dozens of seconds each on standard hardware (Core i9, 16GB RAM).
 - **Accuracy**: >98.3% precision in halogen detection across simulated and experimental LC-MS datasets.
 - **Integration with GNPS**: enhance molecular network annotation in the element dimension by embedding DeepHalo results into GNPS output graphML file
 - **Efficient Dereplication**: significantly higher dereplication rate compared to molecular networking alone in GNPS
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Installation

For EXE Users:

- Double click the exe to launch DeepHalo.
- Available at https://github.com/xieyying/deephalo/releases/tag/DeepHalo_V1.0.0)
and <https://pan.baidu.com/s/1Bxr1aEGl7GDZ74XHBdA1IA?pwd=tyg1>

For WHL Users:

Prerequisites

- Python 3.10 (Verify with `python --version`).

Installation Methods

From PyPI

```
pip install deephalo
```

From Local Wheel

```
pip install path/to/DeepHalo-xxx.whl
```

From Source

```
git clone https://github.com/xieyying/DeepHalo.git  
cd DeepHalo  
pip install -e .
```

Quickstart

For EXE Users:

Double click the exe to launch DeepHalo.

1. Detect Halogenated Compounds in mzML Files

DeepHalo V1.0.0



A deep learning-integrated workflow for high-throughput discovery of halogenated metabolites from HRMS data

(Note: HRMS data and mzML format needed)

Detect Dereplicate

Input (.mzML file or folder):

Output Directory:

Config File (optional):

Blank Folder (optional):

☐ Overwrite Blank

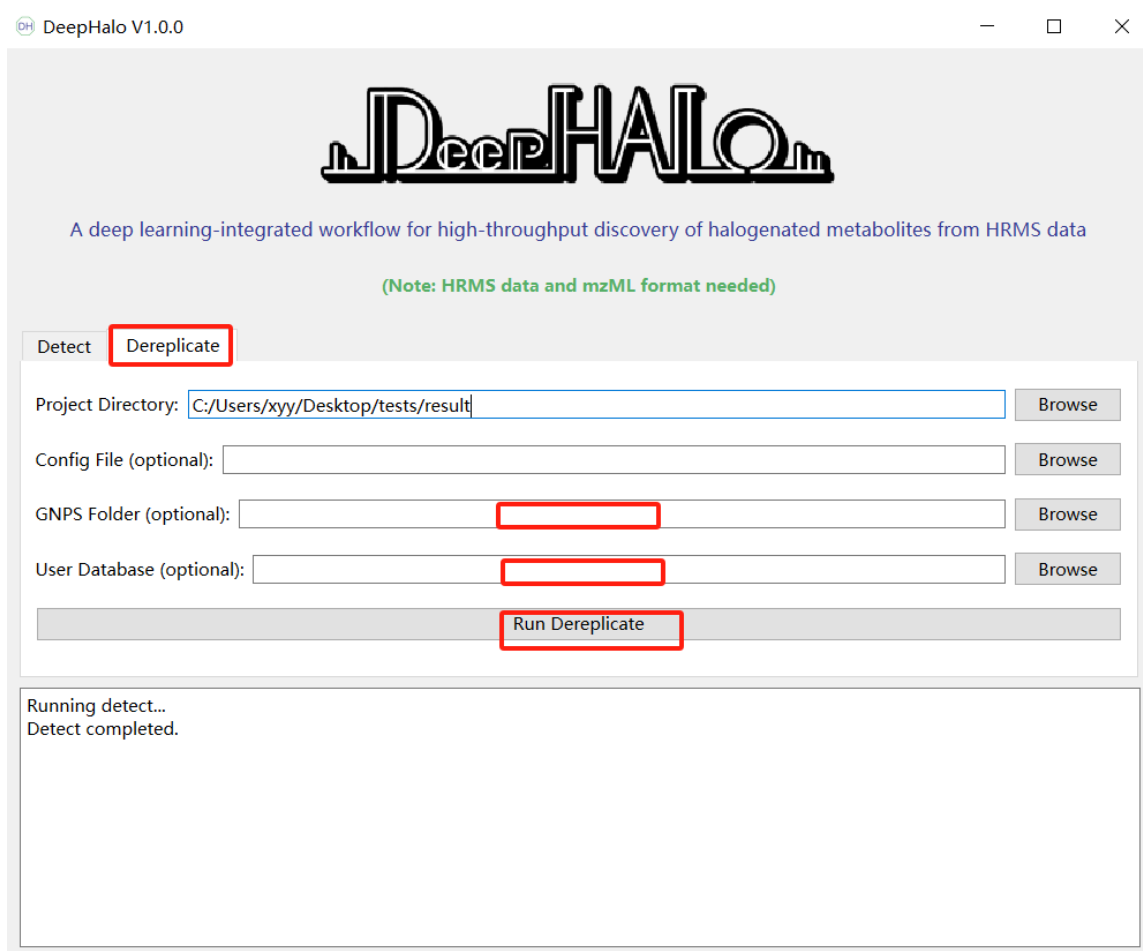
☐ Enable MS2 Extraction (only DDA data supported)

Note: Please use English characters for file and directory names

2. Dereplication with GNPS and/or User Database

For dereplication, you must provide at least one of GNPS results or a User Database.”

(Note: User Database format see below Example DATABASE_FILE)



The screenshot shows the DeepHalo V1.0.0 application window. At the top, there is a logo and a description: "A deep learning-integrated workflow for high-throughput discovery of halogenated metabolites from HRMS data". Below this, a note states: "(Note: HRMS data and mzML format needed)". The interface has two tabs: "Detect" and "Dereplicate", with "Dereplicate" being the active tab. Under the "Dereplicate" tab, there are four input fields: "Project Directory:" (containing "C:/Users/xyy/Desktop/tests/result"), "Config File (optional):", "GNPS Folder (optional):", and "User Database (optional):". Each field has a "Browse" button to its right. The "GNPS Folder" and "User Database" fields are highlighted with red boxes. Below these fields is a "Run Dereplicate" button, also highlighted with a red box. At the bottom of the window, there is a status area that says "Running detect..." and "Detect completed."

For WHL Users:

1. Detect Halogenated Compounds in mzML Files

```
halo detect -i /path/to/mzml_files -o /path/to/output_directory -ms2
```

2. Dereplication with GNPS and/or Custom Database

```
halo dereplicate -o /path/to/output_directory -g /path/to/GNPS_results -ud /path/to/custom_database.csv
```

Command-Line Usage

General Help

```
halo --help # List all commands
```

halo [command] --help # Show options for a specific command (e.g., `halo detect --help`)

Commands

1. Analyze mzML Files

halo detect -i <input_path> -o <project_path> [-c <config_file>] [-b <blank_samples_dir>] [-ms2]

2. Dereplication

halo dereplicate -o <project_path> [-g <GNPS_folder>] [-ud <user_database.csv>]

3. Create Training Dataset

halo create-ds <project_path> [-c <config_file>]

4. Train Model

halo train <project_path> [-c <config_file>] [-m search]

Output Directory Structure

/output_directory

```
├──dereplication
│   ├── Demo_data_1_feature.csv
│   └── Demo_data_2_feature.csv
├──result
│   ├── config.toml
│   ├── error_files.txt
│   ├──halo
│   │   ├── Demo_data_1_feature.csv
│   │   ├── Demo_data_1_scan.csv
│   │   ├── Demo_data_2_feature.csv
│   │   └── Demo_data_2_scan.csv
│   └──ms2_output
│       ├── Demo_data_1.mzML
│       └── Demo_data_2.mzML
```

Instruction

The whole DeepHalo analysis process including 4 steps

Step 1: Halogenate Mining

Run the command:

```
halo detect -i <input_path> -o <project_path> [-c <config_file>] [-b <blank_samples_dir>] [-ob] [-ms2]
```

Parameters:

- -i INPUT_PATH (**Required**)
 - Input .mzML file or directory
 - Example (Linux): -i /home/data/ms_files
 - Example (Windows): -i D:\data\ms_files
- -o OUTPUT_DIR (**Required**)
 - Output directory path
 - Example (Linux): -o /home/analysis/output
 - Example (Windows): -o D:\analysis\output
- -c <config_file> (**Optional**)
 - Parameters for LCMS data analysis
 - Example(Linux): -c /home/deephalo/config.toml
 - Example (Windows): -c D:\deephalo\config.toml
- -b <blank_samples_dir> (**Optional**)
 - Blank .mzML file or directory
 - Example (Linux): -b /home/analysis/blank
 - Example (Windows): -b D:\analysis\blank
 - Run blank exclusion during analysis
 - Default: disabled
- -ob (**Optional**)
 - Force regenerate blank feature detection results for blank exclusion
 - Default: disabled
- -ms2 (**Optional**)
 - Enable MS2 data extraction
 - Optional parameter (required for steps 2-4)
 - Default: disabled

Example Usage:

```
halo analyze-mzml -i D:\data\ms_files -o D:\analysis\output -ms2
```

Output Files

- config.toml: Analysis parameters

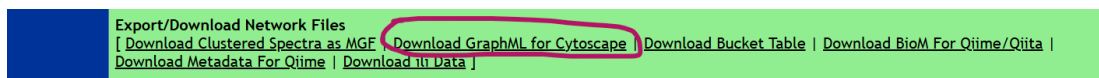
- error_files.txt: Failed mzML files
- halo*: MS1 information
- ms2_output*: MS2 data

Notes

- Results are filtered by H-score
- To disable filtering: Set H-score to 0 in config.toml file
- H-score visible in Cytoscape after dereplication
- For failed files:
 1. Retry analysis
 2. If error persists, reconvert raw data to mzML

Step 2: Molecular Networking analysis employing GNPS

1. Submit MS2 data in ms2_output to GNPS platform
2. Analyze molecular similarity using GNPS platform
3. Download and unzip results



Example GNPS output structure:

```
└─Demo_GNPS_output
  │ METABOLOMICS-SNETS-V2-xxx-main.graphml
  │ params.xml
  └─clusterinfo
    └─clusterinfosummarygroup_attributes_withIDs
      └─gnps_molecular_network_graphml
        └─result_specnets_DB
```

Step 3: Dereplication

Run the command:

```
halo dereplicate -o PROJECT_PATH [-g GNPS_DIR] [-ud DATABASE_FILE]
```

- -o PROJECT_PATH (**Required**)
 - Output directory (same as used in function of 'detect')
 - Example: -o D:\analysis\output
- -g GNPS_DIR (**Optional**)
 - GNPS results directory containing .GraphML file
 - Example: -g D:\analysis\Demo_GNPS_output

- -ud DATABASE_FILE (**Optional**)
 - Custom database file (CSV/JSON)
 - Example: -ud D:\databases\compounds.csv

Output Details

Upon completion of the analysis, the following files and directories will be generated:

1. 'dereplication' Directory

- Contains processed MS1 data in CSV format.
- If a database was provided, compound match information will also be included in the csv file.

2. GraphML File (generated when GNPS results are provided)

- The file name will have the suffix `_adding_DeepHalo_results`.
- This file contains the mean H-score and classification predicted from centroid isotope patterns, along with any database match information if a dereplication database was used.

3. Processed Database File (if a database is provided)

- The file name will include the suffix `_DeepHalo_dereplication_ready_database`.
- This file is formatted for future analyses, significantly reducing processing time in subsequent runs.

Example DATABASE_FILE:

compound_name	formula	Smiles
Compound1	C6H5Cl	<chem>ClC1=CC=CC=C1</chem>
Compound2	C6H4Cl2	<chem>ClC1=CC=C(Cl)C=C1</chem>

Important Notes

1. File Format

- The input file must be in CSV format.

2. Required Columns

- compound_name
- formula

3. Optional (but Recommended) Column

- Smiles (These aid in structure visualization).

4. Additional Requirements

- Ensure that the formula column contains valid molecular formulas.
- The file must be encoded in UTF-8.

Step 4: Visualization in Cytoscape

To visualize the results, open the GraphML file (with the suffix *_adding_DeepHalo_results.graphml) in Cytoscape. The file includes the following annotated variables:

- **H_scoreMean:**
 - **Range:** [0, 1]
 - **Default Threshold:** 0.4, above which the presence of halogen is inferred.
 - **Interpretation:** Higher values indicate a greater likelihood of halogen presence.
 - **Other Factors Affecting the Score:**
 - Sample complexity
 - Mass spectrometer resolution
 - **classification:**
 - **Isotope pattern classifications (0-7):**
 - 0: X-type (mixed/polyhalogenated)
 - 1: Br-type (Br/Cl₃)
 - 2: Cl-type (Cl/Cl₂)
 - 3: Se-type (Selenium compounds)
 - 4: B-type (Boron compounds)
 - 5: Fe-type (Iron compounds)
 - 6: C-type (compounds containing only C, H, O, N, F, P, S, Na)
 - 7: artifact-type (artifact isotope patterns)
 - **Inty_cosine_score:**

Cosine similarity between experimental and theoretical isotope peak intensities. Higher values indicate a greater likelihood of a known compound
 - **Compound_names and Adducts:**

These fields provide the known molecular names and adduct information sourced from a user-provided database, offering essential metadata for compound identification.
 - **Smiles**

Provides the known molecular structure, sourced from either the GNPS library and a user-provided database.
 - **error_ppm**

Represents the mass error between the measured m/z value of the test molecule and that of its corresponding known compound match.
-

Dependencies,

- pandas == 2.0.3
 - numpy == 1.22.0
 - tensorflow == 2.10.1
 - scikit-learn == 1.3.1
 - pyopenms == 3.1.0
 - Full list: See [README.md](#)
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License

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For methodology details and benchmarks, refer to the [GitHub repository](#).